

Singh-Cellular-20713-4-V2

Description

CELLULAR (SDS-Extracted) OVARIAN SAMPLES_V2

Analysis

#1

Report creation date

2026-01-28

Report created with

TARAM v2.11.2

The ZIP file includes two folders:

- **plots:** Contains all plots in PNG, SVG and **interactive, clickable HTML** formats.
- **tables:** Contains all tables in tab-separated format.

Overview of data analysis

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Conclusion

Table-19 is the final table. Enrichment-table-17 is the full GO enrichment table.

-----comments-----

Overall, some differences are clearly visible between early and late time points. The changes of selected proteins appear to correlate with those observed in the soluble sample (ECM).

However, the quality of the final data is sub-optimal. The amounts of material detected by MS in the samples varied quite a lot and this impacted the data, somewhat perturbing the statistical comparisons. At this point we do not know if these differences are due to variability in the resuspension and digestion of the samples or reflect the initial sample amount and composition.

1. Initial result

Table 1

7346 protein groups

Default intensity column:

Quantity

Software version:

Spectronaut 20.4.260109.92449

Analysis date:

23-January-2026

Fasta files:

RefProt_Mus_musculus_20250206.fasta

Universal_Contaminant_Protein_mod_20251210.fasta

Groups and sample identities:

D14	D21	M03	M06	M12	M18
20713	20718	20723	20728b	20733	20740-40ul
20714-40ul	20719b	20724	20729	20734	20741
20715	20720	20725	20730	20735	20742
20716	20721-40ul	20726-40ul	20731	20736	20743
20717	20722	20727	20732	20737	20744

Samples were extracted and digested according to the miST protocol. LC-MS analysis was performed on a TIMS-TOF instrument coupled to an EvoSep system. DIA data were processed by Spectronaut in DirectDia mode.

Note : samples concentrations were not re-quantified before digestion. Amounts injected were adjusted to try to compensate for differences in total MS signal.

2. Log transformation

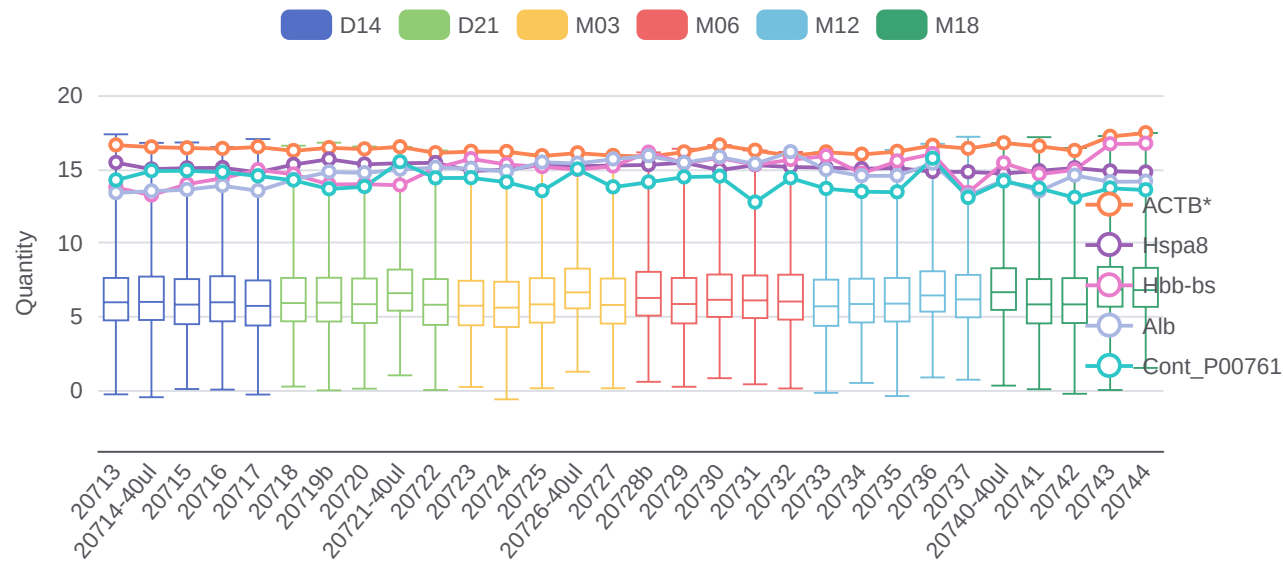
Table 2

7346 protein groups

Log transformation facilitates plotting of data and makes distributions more “normal”, allowing application of standard statistical tests.

3. Boxplot

Overview of samples. Y-axis is usually in log2 scale.



* only the first of multiple gene names is displayed.

Medians are not very well aligned despite global normalization. This suggests overall variability in sample composition.

4. Filter rows

Table 4

7273 protein groups

Parameters:
Remove potential contaminants
Remove `PG.ProteinGroups == *Cont_*`

Protein groups removed: 73

5. Summary

"Nr of valid" = number of quantified proteins.
"NaN" (Not a Number) = missing values.
"Nr of peptides" = sum of "NrOfPrecursorsIdentified"

Name	Group	Min	Max	Median	Nr of valid	Nr of NaN	Nr of peptides
20713	D14	-0.29	17.34	5.95	6164	1109	66008
20714-40ul		-0.49	16.76	5.97	5906	1367	54519
20715		0.07	16.79	5.79	6361	912	66488
20716		0.03	16.48	5.94	6307	966	68087
20717		-0.31	17.01	5.69	6663	610	73210
20718	D21	0.23	16.57	5.89	6432	841	71960
20719b		-0.02	16.77	5.93	6234	1039	69936
20720		0.09	16.52	5.81	6137	1136	64037
20721-40ul		1	16.48	6.57	5017	2256	43365
20722		3.15E-3	16.25	5.77	6220	1053	66788
20723	M03	0.21	15.85	5.71	6880	393	88319
20724		-0.63	16.08	5.58	6774	499	82944
20725		0.12	15.87	5.8	5920	1353	58871
20726-40ul		1.24	15.89	6.61	4800	2473	40929
20727		0.13	15.89	5.77	6011	1262	62492
20728b	M06	0.56	16.12	6.25	5160	2113	52475
20729		0.22	16.36	5.82	6117	1156	64371
20730		0.8	16.3	6.12	5539	1734	53277
20731		0.39	16.16	6.06	5584	1689	51409
20732		0.11	16.15	6	5461	1812	47449
20733	M12	-0.19	15.87	5.67	6224	1049	65598
20734		0.48	16	5.82	6335	938	63386
20735		-0.4	16.29	5.85	6000	1273	60020
20736		0.86	16.69	6.4	5019	2254	46643
20737		0.7	17.18	6.14	5765	1508	54921
20740-40ul	M18	0.3	16.56	6.61	4752	2521	40914
20741		0.06	17.15	5.8	6014	1259	63940
20742		-0.26	16.42	5.8	5678	1595	54731
20743		7.94E-3	17.23	6.81	4398	2875	38151
20744		1.5	17.24	6.76	4585	2688	39992

6. Filter on valid

Retain only proteins quantified in a minimum number of samples.

Table 6

6575 protein groups

Parameters:
Only keep rows where *Quantity* has at least 4 valid value(s) in one group.

Protein groups removed: 698

7. Filter rows

Table 7

6209 protein groups

Parameters:

Only keep *PG.NrOfPrecursorsIdentified (Experiment-wide) > 1*

Protein groups removed: 366

8. Imputation

Table 8

6209 protein groups

Imputation allows calculating fold changes and statistical tests for proteins with missing values. NaN (Not a Number) values are replaced by low-shifted random values, based on the assumption that missing values occur when signals are below detection limits.

Parameters:

Replace missing values in column(s) [Quantity...] by random numbers drawn from a normal distribution:

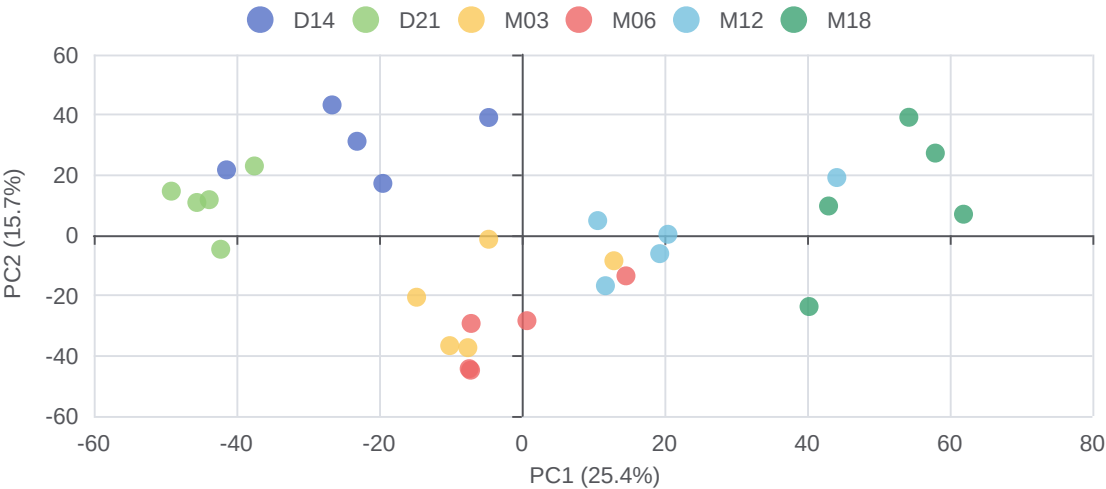
Width: 0.3
Downshift: 1.8
Seed: 1

Nr of imputed values: 23048

Nr of protein groups with imputation: 3200

9. PCA

Examines similarities between samples and groups, needs imputed values.



A partial separation of the time points is visible, mostly along PC1 but also partially in PC2. The two components however only accounts for relatively low percentages of the total variability.

10. t-test

Table 10

Benjamini-Hochberg correction adjusts thresholds for large datasets.

6209 protein groups

Welch's t-test (assumes unequal variances)

Significance threshold: 0.05

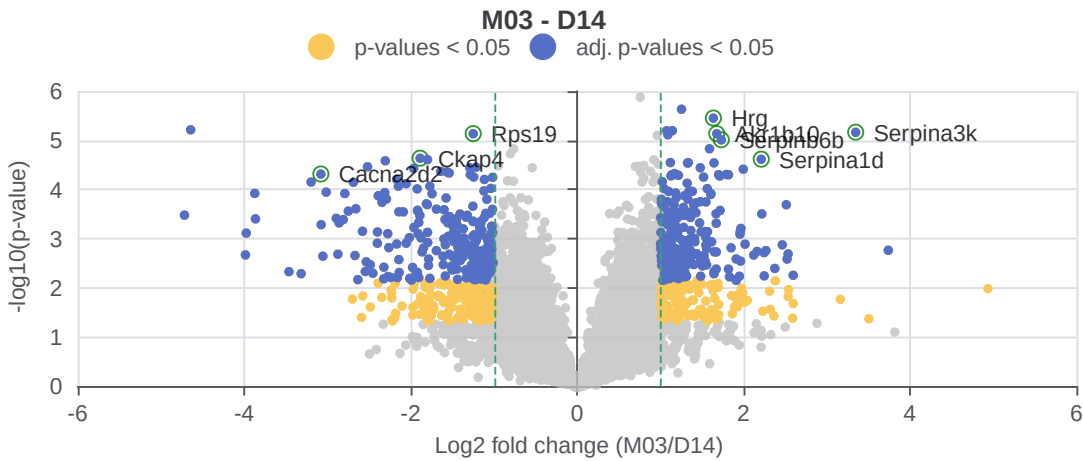
Multiple testing correction: Benjamini-Hochberg (FDR)

Only compute comparisons when there are at least 4 valid (non-imputed) values in one group.

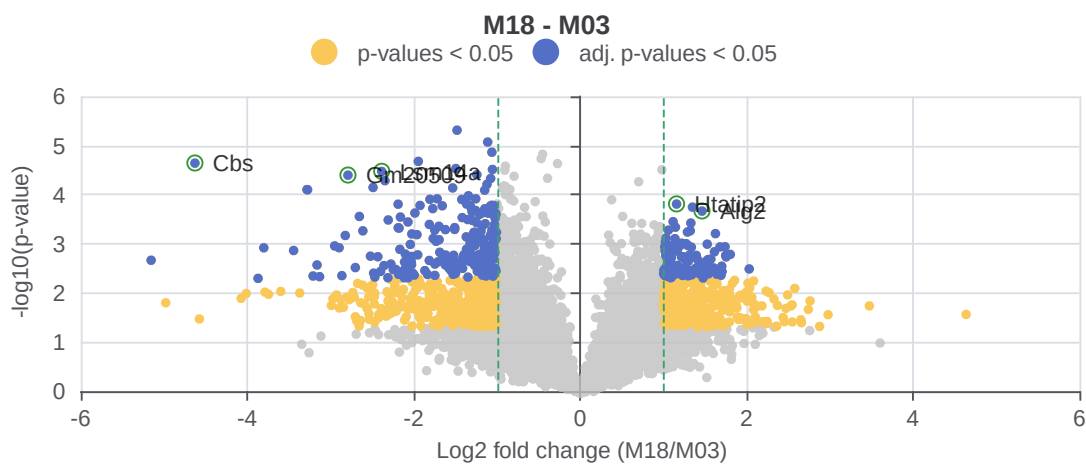
Comparison	Significant	Filter passed
M03 - D14	860	5998
M18 - M03	583	5694

11. Volcano plot

Common way to represent statistical significance (y-axis) and fold change (x-axis). By using $-\log_{10}$ transformation, low p-values are displayed at the top of the plot.



12. Volcano plot



13. Remove imputed

Table 13

Fold changes and p-values calculated with imputed data are kept.

6209 protein groups

Parameters:

Replace imputed values by [NaN].

Nr of replacements:

23048

Nr of protein groups with replacements: 3200

14. 1D Enrichment

Table 14

Table is first ranked by (usually) fold change. Distribution of each annotation term is then evaluated to see if it is enriched toward the top or bottom. FDR-filtered table.

Enrichment table 14

6209 protein groups

Parameters:

Annotation file: mainAnnot.mus_musculus.txt
Annotation info: Mus musculus, PerseusAnnotation from 18-12-23, 2023-12-18, 86221 entries.
Selected annotations: GOMF name, GOBP slim name, KEGG name, GOCC slim name
Selected column(s): t.statistic.M03-D14; t.statistic.M18-M03
Significance threshold: 0.02
Multiple testing correction: Benjamini & Hochberg (FDR)

Selected results:

Column	Type	Name	Size	Score	Adj. p-value	Median
t.statistic.M03-D14	GOMF name	glyceraldehyde-3-phosphate dehydrogenase (NAD+) (non-phosphorylating) activity	7	72.91E-2	16.65E-3	5.118
t.statistic.M03-D14	GOMF name	acyl-CoA dehydrogenase activity	12	61.4E-2	58.55E-4	5.035
t.statistic.M03-D14	GOMF name	retinal dehydrogenase activity	12	86.71E-2	13.48E-6	5.012
t.statistic.M03-D14	GOMF name	alditol:NADP+ 1-oxidoreductase activity	11	90.89E-2	13.29E-6	4.927
t.statistic.M03-D14	KEGG name	Glyoxylate and dicarboxylate metabolism	14	89.56E-2	13.82E-8	4.846
t.statistic.M03-D14	KEGG name	Propanoate metabolism	29	70.7E-2	12.69E-10	4.626
t.statistic.M03-D14	KEGG name	Pyruvate metabolism	35	64.84E-2	10.34E-10	4.24
t.statistic.M18-M03	GOMF name	translation initiation factor activity	41	-49.82E-2	19.59E-7	-2.759
t.statistic.M18-M03	KEGG name	Nucleotide excision repair	25	-57.03E-2	13.02E-6	-2.765
t.statistic.M18-M03	GOMF name	damaged DNA binding	26	-60.65E-2	45.89E-7	-2.848
t.statistic.M03-D14	GOBP slim name	cellular respiration	60	-53.66E-2	80.45E-13	-2.957
t.statistic.M18-M03	GOMF name	phosphatase inhibitor activity	15	-54.12E-2	55.18E-4	-3.068
t.statistic.M18-M03	GOMF name	single-stranded DNA binding	53	-44.67E-2	11.59E-7	-3.086
t.statistic.M03-D14	GOCC slim name	respiratory chain complex IV	15	-53.74E-2	13.36E-4	-3.128
t.statistic.M18-M03	KEGG name	Mismatch repair	13	-71.27E-2	89.78E-6	-3.422
t.statistic.M03-D14	GOMF name	NADH dehydrogenase (quinone) activity	12	-64.85E-2	28.59E-4	-3.634
t.statistic.M03-D14	GOCC slim name	respiratory chain complex III	10	-62.19E-2	27.19E-4	-3.798
t.statistic.M18-M03	KEGG name	DNA replication	27	-77.61E-2	12.21E-11	-3.864
t.statistic.M03-D14	KEGG name	Hematopoietic cell lineage	18	-64.48E-2	25.87E-6	-4.095

15. Remove imputed

Table 15

Fold changes and p-values calculated with imputed data are kept.

6209 protein groups

Parameters:

Replace imputed values by [NaN].

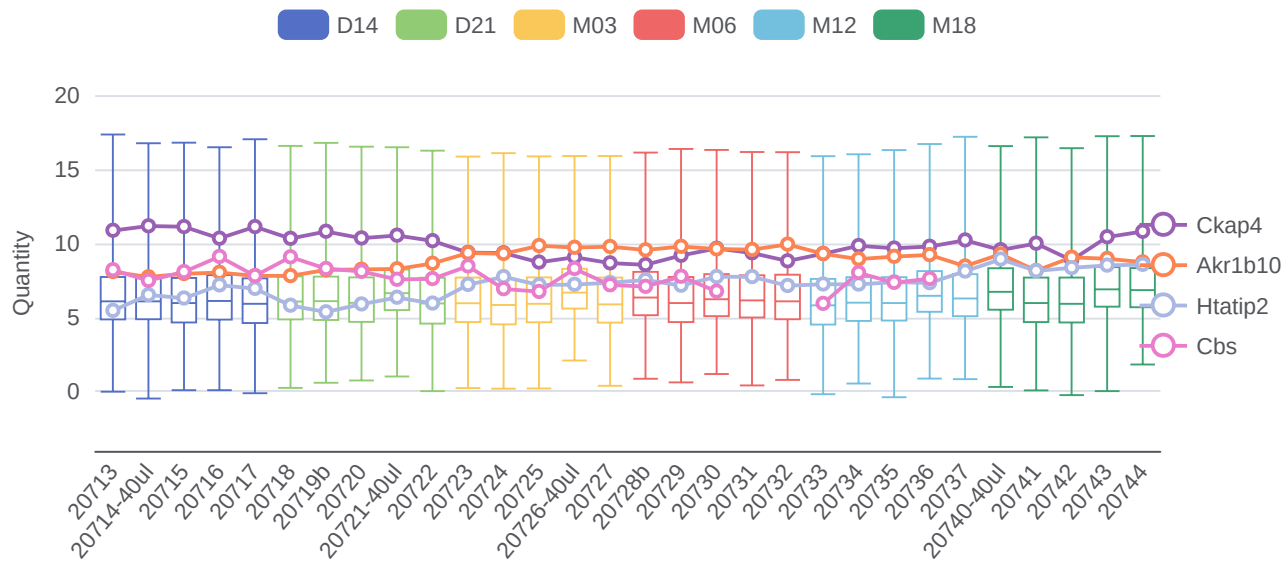
Nr of replacements:

0

Nr of protein groups with replacements: 0

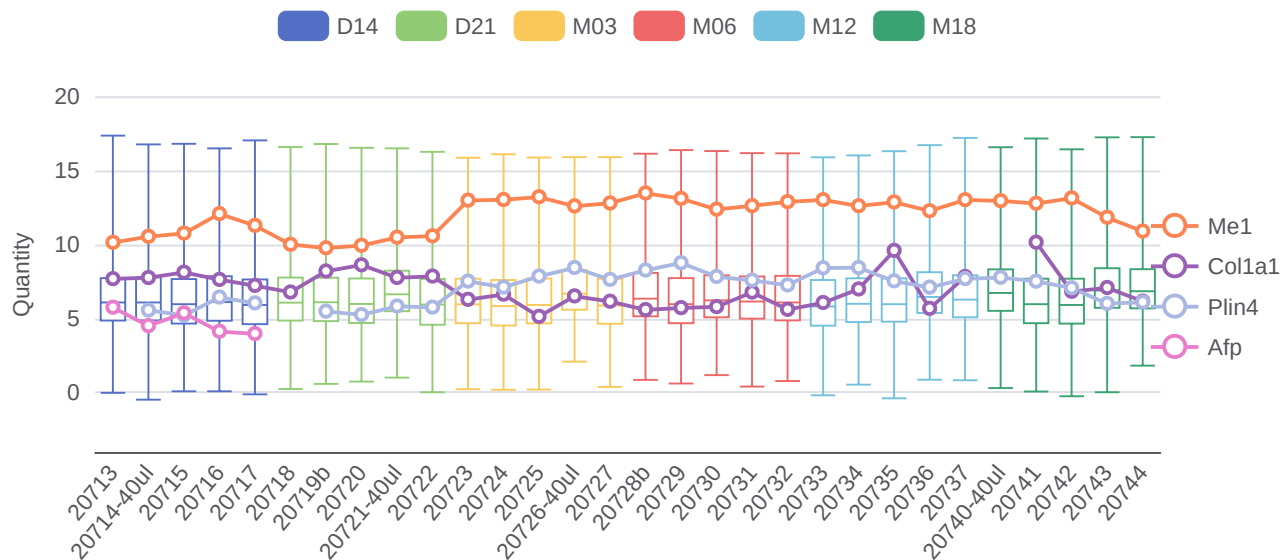
16. Boxplot

Overview of samples. Y-axis is usually in log2 scale.



17. Boxplot

Overview of samples. Y-axis is usually in log2 scale.



Some proteins are shown, that were statistically significantly different in the soluble fraction analysis. They seem to correlate with their whole-cell abundances.

18. Rename headers

Table 18

Add conditions to table headers.

6209 protein groups

19. Order columns

Table 19

Table cleanup.

6209 protein groups

Parameters:

Move default intensity columns [Quantity] first.